Lect 10.2: Differential Equation - I

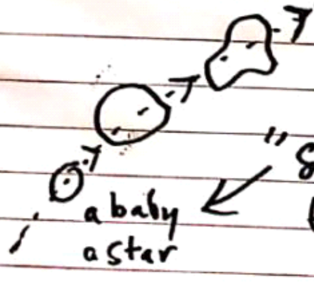
Wednesday, 8 April 2026 5:55 pm

— ODEs Ordinary Differential Equations

All laws of nature are DEs i.e. Differential Equations,

- Laws of Electromagnetism
- Laws of Atomic, Sub-atomic, sub-sub-atomic, world are DEs.
- Laws of Economic Dynamics & Control are DEs. — Macro-economics.
- Laws of Gravity are DEs.
- Laws of Cellular Biology & Dynamics are DEs.

etc



Evolution of Systems.

"System" evolving in space or time, or both.

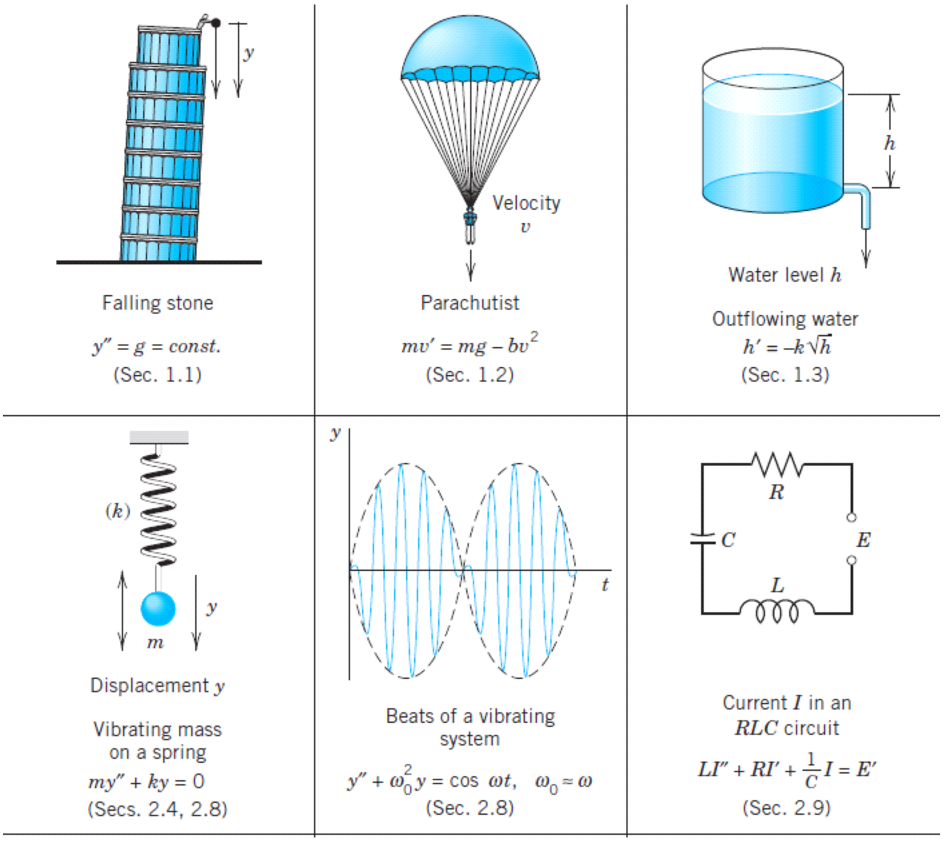
a baby
a star
an economy
etc

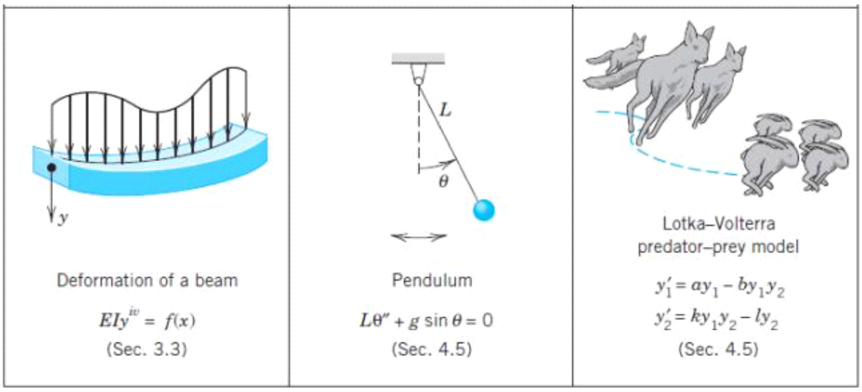
"To solve a DE" means "to integrate the diff. erential equation"

"Integration"

Review of Integration — Techniques of Indefinite Integration:

1 – Algebraic Substitutions
2 – Integration by Parts
3 – Partial Fraction Method
4 – Trigonometric Integrals
5 – Trigonometric Substitution Integrals
6 – Hyperbolic Integrals
etc





Definition 0.1.1

A *differential equation* is an equation that involves one or more derivatives of an unknown function. For a differential equation, the *order* of the differential equation is the highest order derivative that appears in the equation.

One example of a first order differential equation is

$$\frac{dx}{dt} + x = 2\cos t.$$

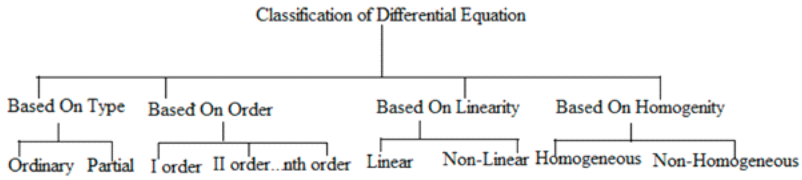
Here x is the *dependent variable* and t is the *independent variable*.

$$\left(\frac{dy}{dt}\right)^2 + y^2 = 1$$

$$\frac{dy}{dt} = \sqrt{1-y^2} \qquad \text{or} \qquad \frac{dy}{dt} = -\sqrt{1-y^2},$$

Classification of Differential Equation

The below given tree is the general classification of differential equation.



$$\frac{d^2y}{dx^2} + \frac{dy}{dx} = 3x \sin y$$

is an ordinary differential equation since it does not contain partial derivatives. While

$$\frac{\partial y}{\partial t} + x\frac{\partial y}{\partial x} = \frac{x+t}{x-t}$$

is a partial differential equation, since y is a function of the two variables x and t and partial derivatives are present.

Order

Another way of classifying differential equations is by order. Any ordinary differential equation can be written in the form

$$F(x, y, y', y'', \ldots, y^{(0)}) = 0$$

by setting everything equal to zero. The order of a differential equation is the **highest** derivative that appears in the above equation.

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} = 3x \sin y$$

$$3y^4y''' - x^3y' + e^{xy}y = 0$$

Differential Equations – Based on Linearity

By linearity, it means that the variable appearing in the equation is raised to the power of one. The graph of linear functions is generally a straight line. For example: $(3x + 5)$ is Linear but $(x^3 + 4x^2)$ is non-linear.

Linear Differential Equation

If all the dependent variables and its entire derivatives occur linearly in a given equation, then it represents a linear differential equation.

Non-Linear Differential Equations

Any differential equation with non-linear terms is known as a non-linear differential equation.

$$3x^2y'' + 2\ln(x)y' + e^x y = 3x\cos x$$

$$4yy''' - x^3y' + \cos y = e^{2x}$$

$$y'' + 2y' + e^y = x$$

Differential Equations – Based on Homogeneity

The linear differential equation of the form,

$fn(x)y^n + \ldots\ldots + f_1(x)y' + f0(x)y = g(x)$
represents a homogeneous differential equation if the R.H.S is zero i.e.,g(x) = 0 , Elseit represents non-homogeneous differential equation if g(x)
$\neq$
0.

$$\frac{d^4y}{dx^4} + x\frac{d^2y}{dx^2} + y^2 = 0$$

$$\frac{d^4y}{dx^4} + x\frac{d^2y}{dx^2} + y^2 = 6x + 3$$

Homogeneous	Non-Homogeneous
x" + x = 0	x" + x = sin(t)
x' + t2x = 0	x' + t2x = t +2t
y''' + xy'' + y = 0	y''' + xy'' + y = 6x + 3
y''sin(x) + 4y +y' = 0	y''sin(x) + 4y +y' = cos(x)

Show that

$$f(x) = x + e^{2x}$$

is a solution to

$$y'' - 2y' = -2.$$

Exercise 0.1.1: *Check that the y given is really a solution to the equation.*

Next, take the *second order differential equation*

$$\frac{d^2y}{dx^2} = -k^2y,$$

for some constant $k > 0$. The general solution for this equation is

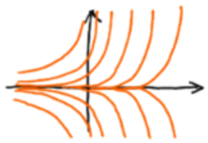
$$y(x) = C_1\cos(kx) + C_2\sin(kx).$$

Example 0.1.3: In equations of higher order, you get more constants you must solve for to get a particular solution. The equation $\frac{d^2y}{dx^2} = 0$ has the general solution $y = C_1x + C_2$; simply integrate twice and don't forget about the constant of integration. Consider the initial conditions $y(0) = 2$ and $y'(0) = 3$. We plug in our general solution and solve for the constants:

GENERAL SOLUTION

Diff. Eq. : $\frac{dy}{dt} = 2y$

General Solution : $y = Ce^{2t}$

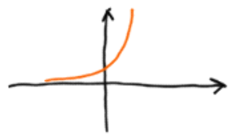


Includes all possible solutions (and no functions that are not solutions)

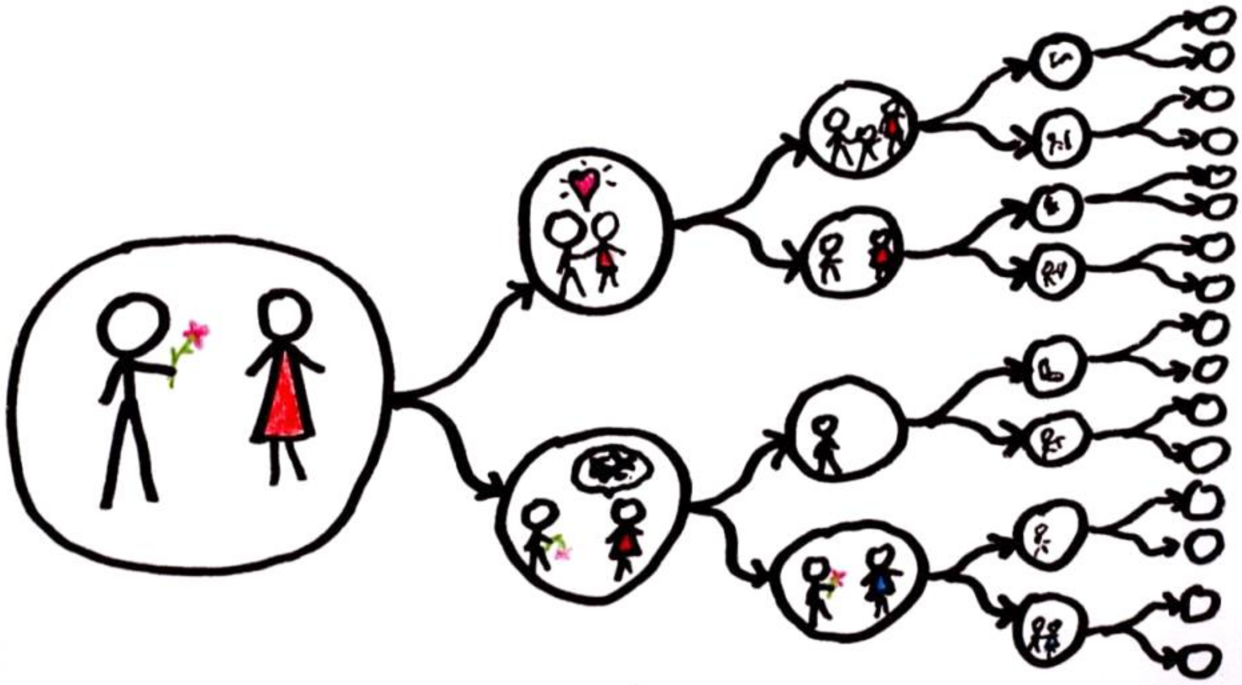
PARTICULAR SOLUTION

Diff. Eq. : $\frac{dy}{dt} = 2y$

Particular Solution : $y = e^{2t}$ $(C = 1)$



One individual solution whose graph is a single curve (an integral curve)



Exercise 0.1.4: *Show that $x = e^{4t}$ is a solution to $x''' - 12x'' + 48x' - 64x = 0$.*

Exercise 0.1.5:* *Show that $x = e^{-2t}$ is a solution to $x'' + 4x' + 4x = 0$.*

Exercise 0.1.6: *Show that $x = e^{t}$ is not a solution to $x''' - 12x'' + 48x' - 64x = 0$.*

Exercise 0.1.7: *Is $y = \sin t$ a solution to $\left(\frac{dy}{dt}\right)^2 = 1 - y^2$? Justify.*

Exercise 0.1.8:* *Is $y = x^2$ a solution to $x^2y'' - 2y = 0$? Justify.*